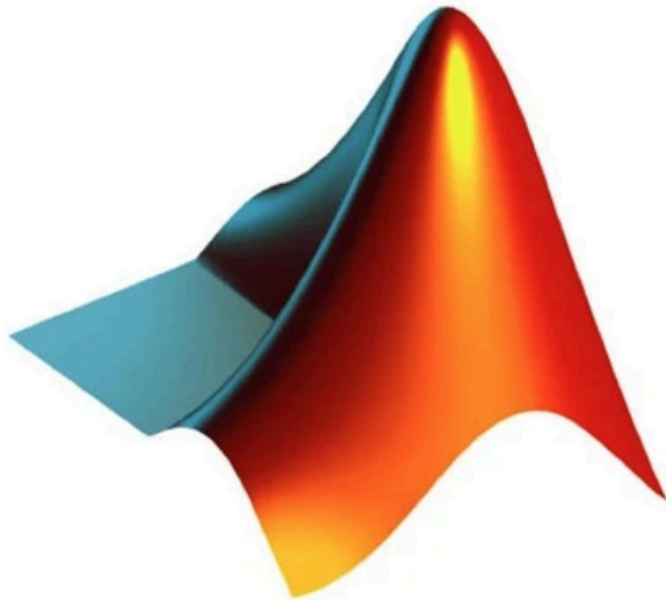
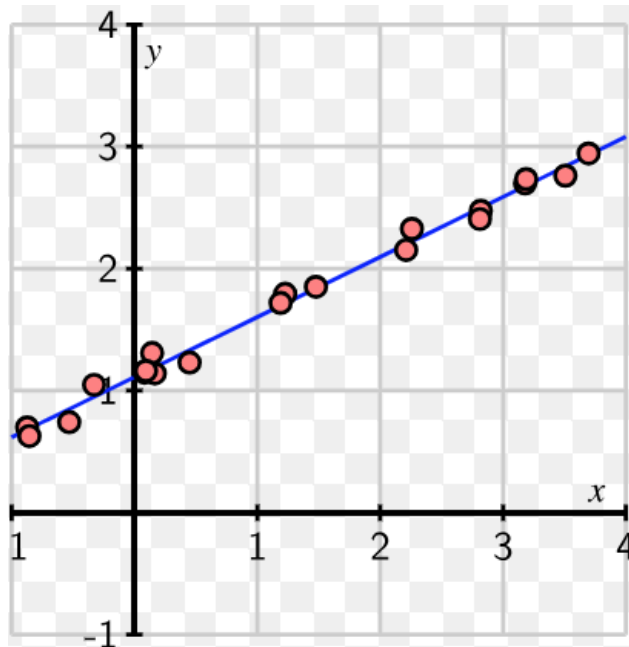


MATH 284 MATLAB Project #5 - Orthogonality and Least Squares Studio



MATLAB®



Project Objective

Develop an interactive MATLAB application that allows users to investigate orthogonality, construct orthogonal bases, perform projections, solve least-squares problems, and explore how these ideas are used in machine learning.

In contrast to the previous projects, this one emphasizes **approximations** rather than exact solutions.

Concepts And Textbook Chapters/Sections Covered

Chapter 6: Orthogonality and Least Squares

- 6.1 Inner Product, Length, and Orthogonality
 - 6.2 Orthogonal Sets
 - 6.3 Orthogonal Projections
 - 6.4 The Gram–Schmidt Process
 - 6.5 Least-Squares Problems
 - 6.6 Machine Learning and Linear Models
-
- Inner products
 - Vector norms
 - Orthogonality
 - Orthogonal sets
 - Orthogonal projections
 - Gram-Schmidt orthogonalization
 - Least-squares approximation
 - Linear regression
 - Machine learning foundations

GUI Layout

Orthogonality & Least Squares Studio		
Input		
Matrix A		
[		]
Vector b		
[		]
Analysis		
[Inner Product]		
[Projection]		
[Gram-Schmidt]		
[Least Squares]		
Results		
Norm		
Angle Between Vectors		
Projection Vector		
Orthogonal Basis		
Least Squares Solution		
Visualization		
Vector Plot		
Projection Animation		
Regression Plot		
Applications		
[Linear Regression]		
[Machine Learning Demo]		
[Approximation Viewer]		

Necessary MATLAB Files For This Project

OrthogonalityStudioApp.m

Responsibilities:

- *Build interface*
- *Handle user interaction*
- *Coordinate calculations*
- *Update visualizations*

OrthogonalityAnalyzer.m

Stores:

- *vectors*
- *inner product*
- *angle*
- *norm*
- *orthogonal*

ProjectionCalculator.m

Computes:

- *Orthogonal projection*
- *Projection error*
- *Orthogonal decomposition*

Visualization:

- *Original vector*
- *Projection onto subspace*
- *Residual vector*

Users immediately see that the residual is perpendicular to the subspace.

GramSchmidtProcess.m

Computes:

- *Orthogonal basis*
- *Orthonormal basis*

Displays:

- *Original Basis*
- *Orthogonal Basis*
- *Orthonormal Basis*

Users can compare the original and transformed basis side by side.

LeastSquaresSolver.m

Computes $Ax \approx b$ using normal equations.

Features:

- *Best-fit solution*
- *Residual vector*
- *Approximation error*

Displays:

- *least squares solution*
- *residual*
- *error*
- *best approximation*

LinearRegressionExplorer.m

Demonstrates:

- *Linear regression*
- *Best-fit line*
- *Prediction*
- *Error metrics*

User can:

- *Enter data*
- *Load sample data*
- *Generate random data*

The best-fit line is calculated automatically.

OrthogonalityVisualizer.m

Responsible for:

- *Vector plots*
- *Projection animations*
- *Gram-Schmidt visualization*
- *Regression graphs*

Necessary MATLAB Methods For This Project

createComponents()

analyzeVectors()

performProjection()

solveLeastSquares()

updatePlots()

clearGUI()

computeInnerProduct()

computeNorm()

computeAngle()

isOrthogonal()

projectVector()

projectionError()

plotProjection()

orthogonalize()

normalizeBasis()

verifyOrthogonality()

solveLeastSquares()

computeResidual()

calculateError()

fitRegression()

predict()

plotRegression()

calculateRMSE()

plotVectors()

plotProjection()

animateGramSchmidt()

plotRegression()

plotResidual()

GUI Functions

- Dot product, norm, angle calculator
- Orthogonality checker
- Interactive projection visualizer
- Orthogonal basis generator
- Least-squares solver
- Linear regression explorer

Pseudocode

START PROGRAM

Create GUI

Initialize

Matrix input

Vector input

Plot panel

Results panel

WHEN Inner Product selected

Compute

Dot product

Norms

Angle

Display results

WHEN Projection selected

Compute projection

Compute residual

Plot vectors

WHEN Gram-Schmidt selected

Orthogonalize vectors

Normalize basis

Display orthonormal basis

WHEN Least Squares selected

Solve least-squares problem

Compute residual

Plot approximation

WHEN Machine Learning selected

Load or input dataset

Fit linear regression model

Display regression equation

Plot best-fit line
